

Question:

Carla is downloading a 200 GB file. Normally she can download 2 GB/minute, but 40% of the way through the download, Windows forces a restart to install updates, which takes 20 minutes. Then Carla has to restart the download from the beginning. How long does it take to download the file?

Reasoning + Answer:

<think> To determine the total time taken to download the file, we need to calculate the time taken to download 60% of the file initially, and then add the time taken for the restart and the subsequent re-download of the entire file. First, we calculate the time taken to download 60% of the 200 GB file at a speed of 2 GB / minute: 60% of 200 GB = 120 GB. Time taken = 120 GB / 2 GB / minute = 60 minutes. Next, we add the 20-minute restart time: 60 minutes + 20 minutes = 80 minutes. Since Carla has to restart from the beginning, the remaining 140 GB (200 GB - 60 GB) will also be downloaded at 2 GB / minute. Time taken to download the remaining 140 GB = 140 GB / 2 GB / minute = 70 minutes. Finally, we add the time taken for the second download: 80 minutes + 70 minutes = 150 minutes. </think><answer> 150 </answer>

Reward score

